

Pholus

AI-Powered Performance Protection for Unity

Version 1.0.0

Documentation & Setup Guide

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1. Overview

Pholus is a Unity Editor tool that uses AI to detect performance anti-patterns in your C# scripts and provides one-click fixes with diff preview. It catches common issues like uncached GetComponent calls, allocations in Update loops, physics calls without layer masks, and 20+ other performance pitfalls.

1.1 Key Features

- AI-Powered Analysis - Uses leading AI models to understand code context, not just pattern matching
- 20+ Detection Rules - Catches the most impactful Unity performance anti-patterns
- One-Click Fix - AI generates fixes with diff preview before applying
- Multi-Provider Support - Works with Claude, OpenAI Codex, Google Gemini, Cursor, and OpenRouter
- Consensus Mode - Run multiple AI providers in parallel for higher accuracy
- Platform-Aware - Adjusts thresholds for Mobile, PC/Console, and VR
- Backup & Undo - Automatic backups before every fix, with full undo support
- Batch Scanning - Analyze entire folders or your whole project at once

2. Requirements

- Unity 2021.3 or later
- Node.js v20 or later (required for AI provider CLI installation)
- npm (included with Node.js)
- An account with at least one supported AI provider (see Section 6: AI Providers)

3. Installation

1. Import the Pholus package into your Unity project.
2. Open Pholus via the Unity menu: Tools > Pholus > Open Pholus
3. The Setup Wizard launches automatically on first use and guides you through configuration.

4. Setup Guide (Step-by-Step Tutorial)

The Setup Wizard walks you through the initial configuration. It launches automatically the first time you open Pholus. You can reopen it anytime from Tools > Pholus > Setup Wizard.

4.1 Step 1: Choose Your AI Provider

Select which AI provider you want Pholus to use for analysis. You only need one provider to get started. You can change your provider at any time in Settings.

Provider	Subscription	Notes
Claude Code	Claude Pro or Max	Recommended. Best accuracy, prompt caching.
OpenAI Codex	ChatGPT Plus	Solid alternative for OpenAI users.
Google Gemini	Free (1000/day) or API key	Free tier available.
Cursor	Free (limited) or Pro	Beta. Requires Cursor agent.
OpenRouter	Pay-per-use	Alpha. Access 100+ models.

Tip: Claude Code is recommended for the best analysis accuracy and prompt caching support.

4.2 Step 2: Install Prerequisites

Pholus checks that Node.js v20+ and npm are installed on your system. If they are not found, the wizard provides download links and platform-specific instructions.

Once prerequisites are confirmed, Pholus installs your chosen provider's CLI tool automatically via npm. For example, for Claude Code it runs: `npm install -g @anthropic-sdk/claude-code`

Installation commands for each provider:

Provider	CLI	Install Command
Claude Code	claude	<code>npm install -g @anthropic-sdk/claude-code</code>
OpenAI Codex	codex	<code>npm install -g @openai/codex</code>
Google Gemini	gemini	<code>npm install -g @google/gemini-cli</code>
Cursor	cursor-agent	<code>curl https://cursor.com/install -fsSL bash</code>
OpenRouter	openrouter	<code>npm install -g @letuscode/openrouter-cli</code>

4.3 Step 3: Authenticate

Authenticate with your chosen provider so Pholus can make API calls on your behalf.

- Claude: Opens a terminal and runs 'claude /login'. Follow the browser prompts to log in.
- Codex: Opens a terminal running 'codex' which triggers interactive login.
- Gemini: Opens a terminal running 'gemini' which auto-authenticates with your Google account.
- Cursor: Verify your Cursor agent is installed and authenticated via 'cursor-agent status'.
- OpenRouter: Enter your API key directly in the Pholus UI. Get a key at openrouter.ai/keys.

After authentication, Pholus runs a verification test to confirm the connection works before proceeding.

4.4 Step 4: Verification and Completion

The wizard confirms that everything is configured correctly. You will see a success message indicating that

Pholus is ready to use. Click 'Finish' to close the wizard and start analyzing code.

If you ever need to reconfigure, go to Tools > Pholus > Setup Wizard to restart the process.

5. Getting Started

5.1 Quick Start

1. Open Pholus from the menu: Tools > Pholus > Open Pholus
2. Drag and drop a C# script into the script list, or click 'Add' to browse for one.
3. Click 'Analyze'.
4. Review the results: issues are listed with severity, explanation, and a 'Fix' button.
5. Click 'Fix' on any issue to generate a fix, review the diff, and apply it.

5.2 Main Window Layout

The Pholus window is organized into these areas:

- Header - Pholus title, token usage (Claude only), provider status, and settings button.
- Mode Selector - Switch between Scripts, Folder, and Project analysis modes.
- Script List - In Scripts mode, shows selected scripts with Add/Remove controls and drag-and-drop.
- Action Button - 'Analyze', 'Scan Folder', or 'Scan Project' depending on the mode.
- Results Area - Detected issues grouped by certainty, with severity colors and fix buttons.
- Footer - 'Undo Last Fix' button appears when a fix has been applied.

5.3 Using the Demo Scripts

Pholus includes example scripts in the Demo/ folder that contain intentional performance issues. These are perfect for testing your setup and learning what Pholus detects.

Try it: Drag any demo script into Pholus and click Analyze to see the tool in action.

6. AI Providers

Pholus supports five AI providers. You only need one to get started, but you can switch between them or use multiple providers simultaneously with Consensus Mode.

6.1 Claude Code (Recommended)

- Best for: Highest accuracy analysis and fixes

- CLI command: claude
- Requires: Claude Pro or Max subscription
- Prompt Caching: Supported. Saves up to 90% on repeated analyses. Cache valid for 5 minutes.
- Token Display: Shows real-time token usage with cache statistics in the Pholus header.

6.2 OpenAI Codex

- CLI command: codex
- Requires: ChatGPT Plus subscription
- Auth file: ~/.codex/auth.json (created after first login)

6.3 Google Gemini

- CLI command: gemini
- Free tier: 1000 requests/day with a Google account
- Also supports paid API key for higher limits
- Auth file: ~/.gemini/settings.json or ~/.gemini/credentials.json

6.4 Cursor (Beta)

- CLI command: cursor-agent
- Requires: Cursor account (free limited usage, or Cursor Pro)
- Windows users: Requires WSL (Windows Subsystem for Linux). Install with: wsl --install
- Mac/Linux: Native execution supported

6.5 OpenRouter (Alpha)

- CLI command: openrouter
- Pay-per-use pricing with access to 100+ AI models
- Model format: provider/model-name (e.g., anthropic/claude-3.5-sonnet)
- API key from: openrouter.ai/keys (entered directly in Pholus UI)

6.6 Switching Providers

Change your active provider anytime in Settings > AI Provider. Each provider maintains its own model selection independently, so switching back preserves your previous configuration.

6.7 Model Selection

- Use '(CLI Default)' to let the provider automatically choose the best model.
- Click 'Refresh' in settings to discover available models for your provider.

- Click 'Add' to manually enter a model name not in the discovered list.
- Click 'Remove' to delete custom models you have added.

7. Analysis Modes

7.1 Scripts Mode (Default)

Analyze individual C# scripts:

- Add scripts via the 'Add' button or by dragging and dropping files into the list.
- Remove scripts with the X button next to each entry.
- Click 'Analyze' to scan all selected scripts.
- Results are cached so you can re-inspect them without re-analyzing.

7.2 Folder Mode

Scan an entire folder recursively:

- Click 'Browse' to select a folder.
- Click 'Scan Folder' to analyze all .cs files within it.
- Progress shows the current file being analyzed and overall completion.
- Results list all files with a summary score.

7.3 Project Mode

Scan your entire Unity project:

- Analyzes all scripts under Assets/.
- Automatically excludes Plugins and Editor folders.
- Click 'Scan Project' to begin.
- Best for comprehensive audits before release.

8. Understanding Results

8.1 Issue Certainty Levels

Issues are categorized by how confident the AI is that they are real problems:

Level	Meaning	Example
Definite	Almost certainly a performance issue	GetComponent inside Update()
Contextual	Depends on usage context	Find() in a rarely called method

Suggestion	Optional improvement	Code style or minor optimization
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8.2 Severity Ratings

Each issue is assigned a severity level:

Severity	Score Impact	Description
Critical	-15 points	Must fix. Causes significant frame drops.
High	-10 points	Should fix. Noticeable performance impact.
Medium	-5 points	Consider fixing. Minor but measurable impact.
Low	-2 points	Nice to fix. Minimal impact.

8.3 Score Calculation

Each script receives a score from 0 to 100, starting at 100 with deductions per issue based on severity. A score of 100 means no issues were found.

8.4 Issue Details

Each detected issue includes:

- Title - Short description of the problem.
- Line Number - Where in your code the issue occurs.
- Code Snippet - The specific problematic code.
- Explanation - Why this is a performance issue and how it affects your game.
- Impact - Estimated performance cost (e.g., '~0.3ms per frame').
- Why I Might Be Wrong - Conditions where this might be acceptable (toggle in Settings).

8.5 What Pholus Detects

Pholus detects 20+ performance anti-patterns, including:

Issue	What It Catches
Uncached GetComponent	GetComponent<T>() in Update/FixedUpdate/LateUpdate
Uncached Camera.main	Camera.main lookup without caching the reference
Uncached Find	Find()/FindObjectOfType() in frequently called methods
String Concatenation	String += or Concat in Update causing allocations
LINQ in Update	Where(), OrderBy(), FirstOrDefault() allocations
Allocations in Update	new List<>(), new Array[] in Update loops
Empty Unity Methods	Unused OnEnable/Start that Unity still calls
Physics No Layer Mask	Raycast/SphereCast without layer mask filtering
SendMessage Usage	SendMessage/BroadcastMessage reflection overhead
Tag Comparison	tag == "string" instead of CompareTag()

Debug.Log in Update	Debug.Log in performance-critical loops
Animator String Hash	SetTrigger("string") instead of using hash IDs
Coroutine Allocations	new WaitForSeconds() instead of cached instance
Resources.Load	Runtime asset loading in tight loops

8.6 Confidence Modifiers

The AI adjusts its confidence based on code context:

Confidence is reduced when:

- Class name suggests a singleton (Manager, Controller, Singleton)
- Code is inside `#if UNITY_EDITOR` or `[Conditional("DEBUG")]`
- Method is event-driven (OnClick, OnTrigger, OnEnable)
- Collections are small and fixed-size (fewer than 20 items)

Confidence is increased when:

- Class name suggests mass instantiation (Enemy, Bullet, Projectile)
- Code is inside a loop (for, foreach, while)
- Multiple issues compound together in the same method
- Code runs in high-frequency methods (Update, FixedUpdate, LateUpdate)

9. Fixing Issues

9.1 Fix Workflow

1. Click the 'Fix' button on any detected issue.
2. Pholus sends the issue context and your original code to the AI provider.
3. The AI generates a complete fix for the issue.
4. A diff preview window shows exactly what will change: green = additions, red = removals.
5. Click 'Apply' to accept, 'Apply & Don't Ask' to skip previews, or 'Cancel' to discard.

9.2 Backup System

When enabled (on by default), Pholus creates a backup before every fix:

- Location: A `.pholus-backup/` folder is created next to your script.
- Naming: Backups are named `ScriptName.cs.yyyyMMdd_HH:mm:ss.bak` with timestamps.
- Limit: Configurable from 1 to 50 (default 10). Oldest backups auto-deleted when limit exceeded.

9.3 Undo

- After applying a fix, an 'Undo Last Fix' button appears in the footer bar.
- Clicking it restores the file from the most recent backup.
- Up to 50 undo actions are kept per session.

9.4 Auto-Apply Mode

In Settings > Preferences, you can enable 'Auto-Apply Fixes' to skip the diff preview and apply fixes immediately when you click 'Fix'. Backups are still created and undo remains available.

10. Consensus Mode

10.1 What Is Consensus Mode?

Consensus Mode runs your analysis through multiple AI providers simultaneously. A designated 'Director' provider then reviews all opinions and decides which issues are genuinely valid. This reduces false positives and increases confidence in results.

10.2 How It Works

1. Analyzers: Two or more providers each independently analyze your code in parallel.
2. Director Review: The Director receives all opinions and evaluates each issue.
3. Final Verdict: Valid issues are kept with agreement tags (e.g., [2/3] means 2 of 3 providers found the issue). Dismissed issues shown separately with reasoning.

10.3 Setting Up Consensus Mode

1. Go to Settings > Consensus Mode.
2. Enable 'Multi-Provider Consensus'.
3. Check at least 2 authenticated providers as Analyzers.
4. Select a Director provider (Claude is recommended for best judgment accuracy).

10.4 Provider Status Icons

Icon	Meaning
Green check	Authenticated and ready to use
Orange warning	CLI installed but not yet authenticated
Gray X	CLI not installed

10.5 When to Use Consensus Mode

- Before major releases, for highest confidence results.
- When you want to minimize false positives on critical code paths.
- When analyzing performance-critical systems where accuracy matters most.
- When you want multiple AI perspectives on contextual issues.

11. Settings Reference

Open Settings via the gear icon in the top-right corner of the Pholus window.

11.1 AI Provider

Setting	Description
Active Provider	Which AI provider to use for analysis.
Provider Status	Shows current connection and authentication status.
Model Selection	Choose a specific model or use CLI default.
Refresh	Re-discover available models for the selected provider.
Login / Relogin	Open terminal for provider authentication.

11.2 Analysis Options

Setting	Description	Default
Target Platform	Mobile / PC-Console / VR. Adjusts thresholds.	PC/Console
Show 'Why I Might Be Wrong'	Display AI reasoning for contextual issues.	Off
Group by Certainty	Group as Definite > Contextual > Suggestions.	On

11.3 Platform Thresholds

Platform	Frame Budget	Strictness
Mobile	16ms (60 FPS)	Strict. Every allocation matters.
PC/Console	16ms (60 FPS)	Standard. More memory headroom.
VR	11ms (90 FPS)	Very strict. Stutter highly noticeable.

11.4 Fix Options

Setting	Description	Default
Create Backup Before Fix	Save a backup copy before modifying.	On
Max Backups to Keep	1-50 per script. Oldest auto-deleted.	10

11.5 Preferences

Setting	Description	Default
Auto-Apply Fixes	Skip diff preview, apply immediately.	Off

11.6 Logging

Setting	Description	Default
Log Debug Messages	Show info logs in Unity Console.	Off
Log Warnings	Show warning messages.	On
Log Errors	Show error messages.	On

11.7 Cache Management

Action	What It Does
Clear Cache	Removes cached analysis results. Does not affect backups.
Clear Models	Resets discovered and custom model lists.
Reset to Defaults	Restores all settings to factory defaults.

12. Demo Scripts

Pholus includes example scripts in the Demo/ folder that contain intentional performance issues. Use these to test your setup, learn what Pholus detects, and see how fixes are generated.

Script	Issue Demonstrated
UncachedComponents.cs	GetComponent in Update/FixedUpdate without caching
LinqInUpdate.cs	LINQ query allocations inside Update
PerformanceAntiPatterns.cs	Multiple anti-patterns combined
BadCoroutines.cs	WaitForSeconds allocations in coroutines
SendMessageUsage.cs	SendMessage/BroadcastMessage overhead
UpdateAllocations.cs	Various allocation patterns in Update
ExpensiveLookups.cs	Find/GetComponent patterns in hot paths
PhysicsNoLayerMask.cs	Raycast without layer mask optimization
TagComparison.cs	String tag comparison instead of CompareTag
DebugInUpdate.cs	Debug.Log in performance-critical methods
AnimatorStrings.cs	String-based animator parameter calls

Try it: Drag any demo script into Pholus and click 'Analyze' to see the tool in action.

13. Troubleshooting

13.1 CLI Not Detected

Symptom: Provider shows 'Not installed' in Settings.

Solutions:

1. Verify Node.js v20+ is installed by running 'node --version' in your terminal.
2. Install the provider CLI manually (see Section 4.2 for commands).
3. Restart Unity after installing the CLI.
4. On Windows, ensure npm's global bin directory is in your system PATH.
5. Re-run the Setup Wizard: Tools > Pholus > Setup Wizard.

13.2 Authentication Failed

Symptom: Provider shows 'Not authenticated' after login attempt.

Solutions:

1. Run the login command directly in your terminal to see full error output.
2. Verify your subscription is active with the provider.
3. Check for credential files (see Section 6 for each provider).
4. Try the 'Relogin' button in Pholus settings.

13.3 Analysis Returns No Results

Symptom: Analysis completes but shows no issues.

Solutions:

1. Your code may genuinely have no issues (score: 100). This is a good thing!
2. Try analyzing one of the included demo scripts to verify the tool is working.
3. Check the Unity Console for error messages.
4. Verify your AI provider is responding by checking your network connection.

13.4 Model Unavailable Error

Symptom: Error message about a model that cannot be used.

Pholus handles this automatically by resetting to the CLI default model and retrying the analysis. If the error persists, click 'Refresh' in the model settings to discover currently available models.

13.5 Fix Not Applying

Symptom: Fix appears to succeed but code does not change.

Solutions:

1. Close the script in your IDE/editor before applying the fix.
2. Check the Unity Console for error messages.

3. Verify the script file is not read-only.
4. Try generating the fix again. The AI occasionally produces incomplete output.

13.6 Cursor on Windows

Symptom: Cursor provider does not work on Windows.

The Cursor agent requires WSL (Windows Subsystem for Linux). Open an administrator PowerShell and run 'wsl --install', then restart your computer. After WSL is installed, the Cursor provider will work.

14. Contact and Support

For questions, bug reports, and feature requests:

- Email: codeturion@gmail.com
- Discord: [codeturion](#)

You can also find the Contact Us section in Pholus Settings for quick access.